Panduan Checkpoint Pertemuan 9 — Perintah per VM

Semua perintah di bawah disusun berurutan. Label di awal setiap blok menunjukkan **di mana** perintah dijalankan:

Label	Arti
`[HOST]`	Windows (PowerShell/git-bash), tempat VirtualBox
`[TARGET]`	VM `PBL-K4-01-Target-Ubuntu` (192.168.4.5)
`[KALI]`	VM `PBL-K4-02-Attacker-Kali` (192.168.4.100)
`[ONION]`	VM `PBL-K4-03-Monitoring-Onion` (192.168.4.200)

Kredensial lab: `iotadmin` / `123` (Target), `kali` / `kali` (Kali), `analyst` / `cyberops` (Onion).

Checkpoint yang harus tercapai

#	Checkpoint	Bukti yang disimpan
1	Tiga VM menyala, semua di `pbl-kel4`	screenshot tiga VM / `VBoxManage showvminfo`
2	Ketiga node saling ping	keluaran `ping` (0% loss)
3	Dashboard KABITA terbuka di browser	screenshot `http://192.168.4.5/`
4	API mengembalikan data PostgreSQL	`curl /api/ringkasan`
5	Sensor Onion melihat trafik lab	`tcpdump` di Onion
6	Baseline traffic terekam	`baseline_kabita.pcap` + metrik
7	Hardening awal aktif	`ufw status`, `fail2ban-client status sshd`
8	Slide + laporan terunggah	`git push` sukses, file ada di GitHub

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0. [HOST] Nyalakan lab

```
VBM="/c/Program Files/Oracle/VirtualBox/VBoxManage.exe"
"$VBM" startvm "PBL-K4-01-Target-Ubuntu" --type headless
"$VBM" startvm "PBL-K4-02-Attacker-Kali"  --type gui
"$VBM" startvm "PBL-K4-03-Monitoring-Onion" --type gui
```

Cek adaptor lab sudah benar (harus `intnet pbl-kel4`):

```
"$VBM" showvminfo "PBL-K4-01-Target-Ubuntu" | grep -i "NIC 1"
"$VBM" showvminfo "PBL-K4-03-Monitoring-Onion" | grep -iE "NIC 1|Promisc"
```

Snapshot bersih sebelum eksperimen:

```
"$VBM" snapshot "PBL-K4-01-Target-Ubuntu" take "pra-skenario" --description "sebelum uji"
```

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1. [TARGET] Pastikan layanan KABITA hidup

```
systemctl is-active kabita postgresql fail2ban ufw
curl -s -o /dev/null -w 'dashboard: %{http_code}\n' http://127.0.0.1/
curl -s http://127.0.0.1/api/ringkasan
```

Harapan: `active` ×4, `dashboard: 200`, dan JSON `{"jumlah_kambing":"6", ...}`.

Kalau `kabita` mati:

```
sudo journalctl -u kabita -n 30 --no-pager
sudo systemctl restart kabita && sleep 3 && systemctl is-active kabita
```

Kalau data kosong / ingin reset data (data lama hilang):

```
# [HOST] dari folder repo
python scripts/docs/deploy_kabita.py --db-only
```

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2. [TARGET] Cek jaringan & port

```
ip -4 -o addr show enp0s3            # harus 192.168.4.5/24
ss -tulpn | grep LISTEN              # 22, 80 (0.0.0.0), 5432 (127.0.0.1)
sudo ufw status verbose              # active, deny incoming, allow 22 & 80
```

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3. [KALI] Siapkan IP lab & uji koneksi

Login TTY: `Ctrl+Alt+F3`, lalu user `kali`, password `kali`.

```
sudo nmcli con add type ethernet ifname eth0 con-name pbl-lab ip4 192.168.4.100/24
sudo nmcli con up pbl-lab
ip -4 -o addr show eth0              # harus 192.168.4.100/24
ping -c3 192.168.4.5
curl -s -o /dev/null -w 'HTTP %{http_code}\n' http://192.168.4.5/
curl -s http://192.168.4.5/api/kambing | head -c 200
```

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4. [ONION] Pastikan sensor melihat trafik lab

Login: user `analyst`, password `cyberops`.

```
sudo ip link set eth0 promisc on arp on
sudo ip addr add 192.168.4.200/24 dev eth0
ping -c3 192.168.4.100
sudo tcpdump -i eth0 -nn host 192.168.4.100 -c 10
```

Sambil `tcpdump` berjalan, jalankan di **[KALI]**: `ping -c5 192.168.4.5`.

Harapan: 10 baris paket muncul di Onion.

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5. [TARGET] Rekam baseline traffic normal

Terminal 1 — **[TARGET]**:

```
sudo tcpdump -i enp0s3 -nn -w ~/baseline.pcap -c 900
```

Terminal 2 — **[KALI]**:

```
for i in $(seq 1 15); do
  curl -s -o /dev/null http://192.168.4.5/
  curl -s -o /dev/null http://192.168.4.5/api/kambing
  ping -c1 -W1 192.168.4.5 >/dev/null
  sleep 2
done
```

Setelah loop selesai, **[TARGET]**:

```
sudo tcpdump -r ~/baseline.pcap -nn | wc -l
sudo cp ~/baseline.pcap /tmp/baseline_kabita.pcap
```

Simpan ke repo dari **[HOST]** (lihat bagian 7).

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6. [TARGET] Terapkan / verifikasi hardening awal

```
# patch
sudo apt update && sudo apt -y full-upgrade
```

```
# minimalisasi service
for s in ModemManager snapd fwupd multipathd upower udisks2; do
  sudo systemctl disable --now "$s" 2>/dev/null
done

# firewall
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp comment 'admin ssh'
sudo ufw allow from 192.168.4.0/24 to any port 80 proto tcp comment 'dashboard KABITA'
sudo ufw --force enable
sudo ufw status verbose

# brute force
sudo systemctl enable --now fail2ban
sudo fail2ban-client status sshd
```

Simpan bukti (jalankan di **[TARGET]**, hasilnya disalin ke repo pada bagian 7):

```
{
  echo "### Sistem";        lsb_release -ds; uname -r
  echo "### Port";          ss -tulpn | grep LISTEN
  echo "### Service";       systemctl list-units --type=service --state=running --no-pager --no-legend | awk
'{print $1}' | sort
  echo "### ufw";           sudo ufw status verbose
  echo "### fail2ban";      sudo fail2ban-client status sshd
  echo "### KABITA";        systemctl is-active kabita; curl -s -o /dev/null -w '%{http_code}\n'
http://127.0.0.1/
  echo "### upgradable";    apt list --upgradable 2>/dev/null | tail -n +2 | wc -l
} | sudo tee /tmp/after.txt
```

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7. [HOST] Ambil bukti ke repo

```
cd "D:/materi kuliah/Keamanan Siber/PBL"

# pcap baseline
scp -P 2222 iotadmin@127.0.0.1:/tmp/baseline_kabita.pcap \
  docs/blueteam/evidence/baseline/baseline_kabita.pcap

# keluaran hardening
scp -P 2222 iotadmin@127.0.0.1:/tmp/after.txt \
  docs/blueteam/evidence/hardening/after.txt

# metrik (hitung manual dari pcap)
sudo tcpdump -r docs/blueteam/evidence/baseline/baseline_kabita.pcap -nn 2>/dev/null | wc -l
```

Cek cepat isi repo:

```
git status
ls docs/blueteam/evidence/baseline docs/blueteam/evidence/hardening
```

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8. [HOST] Slide, laporan, dan push

```
# slide (HTML -> PDF, 1280x720)
# buka docs/presentation-minggu9/slides.html lalu Print to PDF (lihat README di folder itu)

# laporan
python scripts/docs/md_report_to_docx.py \
  docs/reports/fase1/laporan-fase1.md \
  docs/reports/fase1/Laporan-Fase1-TEK1314-Kelompok4.docx

# commit & push
git add -A
```

```
git commit -m "checkpoint minggu 9: lab KABITA (Nuxt + PostgreSQL), baseline, hardening awal"
git push origin main
```

Screenshot dashboard tanpa GUI

Ubuntu Server tidak memasang desktop, jadi **jangan** mencari Firefox/Chrome di VM Target.

Ada tiga cara yang benar:

Cara 1 — skrip otomatis (paling cepat, dari Windows)

```
# [HOST] sekali saja
pip install paramiko websocket-client

# [HOST] setiap kali butuh bukti baru
python scripts/docs/screenshot_dashboard.py
```

Skrip membuka SSH tunnel `127.0.0.1:8090 -> 127.0.0.1:80`, menjalankan Chrome headless, menyimpan `BUKTI_login.png` dan `BUKTI_dashboard.png` ke `docs/blueteam/evidence/`.

Firewall tidak diubah: traffic masuk lewat SSH, bukan lewat NAT.

Cara 2 — SSH tunnel manual + browser Windows

```
# [HOST] terminal 1
ssh -p 2222 -L 8090:127.0.0.1:80 iotadmin@127.0.0.1     # password: 123

# [HOST] lalu buka di browser Windows
#   http://127.0.0.1:8090        login admin / admin123
#   tekan Win+Shift+S untuk menangkap layar, simpan ke docs/blueteam/evidence/
```

Port 8080 lewat NAT **tidak bisa** dipakai: ufw hanya mengizinkan port 80 dari `192.168.4.0/24`, sehingga request dari host (10.0.3.x) diblokir.

Cara 3 — dari Attacker Node (paling "seperti penyerang melihat")

```
# [KALI] desktop Kali punya Firefox
firefox http://192.168.4.5        # login admin / admin123, lalu tangkap layar
```

Gunakan cara 3 kalau dosen meminta bukti tampilan dari sudut pandang penyerang.

Yang harus kamu kerjakan sendiri (tidak bisa diotomasi)

#	Tugas	Cara
1	**Screenshot dashboard** untuk bukti	Ubuntu Server tidak punya browser. Pakai salah satu cara di bagian **Screenshot dashboard tanpa GUI** di bawah. Hasil: `docs/blueteam/evidence/BUKTI_dashboard.png` (sudah tersedia)
2	**Screenshot sensor Onion**	Jalankan `tcpdump` di Onion sambil ping dari Kali, tangkap terminalnya → `docs/blueteam/evidence/BUKTI_sensor.png`
3	**Push ke GitHub**	Butuh Personal Access Token: GitHub → Settings → Developer settings → Tokens (classic) → Generate, scope `repo`. Lalu `git push` saat diminta user = `rakasyau`, password = token
4	**Unggah slide sebelum sesi**	`docs/presentation-minggu9/slides.pdf` harus sudah ada di GitHub sebelum kelas mulai
5	**Isi nama & NIM di slide/laporan**	Sudah terisi; cek ulang di `slides.pdf` halaman 2 dan laporan halaman 1
6	**Cabut sudo NOPASSWD sebelum penilaian**	Di Target: `sudo rm /etc/sudoers.d/90-pbl-nopasswd` (setelah semua otomasi selesai)

Kalau ada yang gagal

Gejala	Perbaikan
`dashboard: 000` dari Kali	cek `ufw status` di Target; pastikan rule `80 from 192.168.4.0/24` ada
`502`/`500` di dashboard	`sudo journalctl -u kabita -n 50`; biasanya koneksi PostgreSQL
`ping` gagal antar VM	cek NIC 1 ketiga VM = `intnet pbl-kel4`
IP hilang setelah reboot Kali	`sudo nmcli con up pbl-lab`
Onion tidak melihat paket	`sudo ip link set eth0 promisc on arp on`